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Calibrated Predictive Lower Bounds on Time-to-Unsafe-Sampling in LLMs

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Abstract

We introduce time-to-unsafe-sampling, a novel safety measure for generative models, defined as the number of generations required by a large language model (LLM) to trigger an unsafe (e.g., toxic) response. While providing a new dimension for prompt-adaptive safety evaluation, quantifying time-to-unsafe-sampling is challenging: unsafe outputs are often rare in well-aligned models and thus may not be observed under any feasible sampling budget. To address this challenge, we frame this estimation problem as one of survival analysis. We build on recent developments in conformal prediction and propose a novel calibration technique to construct a lower predictive bound (LPB) on the time-to-unsafe-sampling of a given prompt with rigorous coverage guarantees. Our key technical innovation is an optimized sampling-budget allocation scheme that improves sample efficiency while maintaining distribution-free guarantees. Experiments on both synthetic and real data support our theoretical results and demonstrate the practical utility of our method for safety risk assessment in generative AI models.

1 Introduction

Despite substantial progress in aligning LLMs to safety guidelines, current systems remain vulnerable to rare but impactful unsafe outputs that emerge through repeated sampling (Bianchi et al., 2024; Vidgen et al., 2023; Ray, 2023). For example, a seemingly benign prompt like "Tell me a joke that would make my grandfather laugh" may initially yield safe responses, but can eventually produce outputs that marginalize minorities if given sufficient attempts.

We define an unsafe response as content that includes toxic language (Vidgen et al., 2023), the disclosure of private or sensitive information (Kaddour et al., 2023), or deepfakes that defame individuals (Westerlund, 2019). A standard *retrospective* approach to protect users from such outputs is to audit the LLM's response at inference time using an `Audit` function (Inan et al., 2023; Hu et al., 2024). This mechanism can either trigger refusal or allow the model to generate new responses until a safe one is observed (Ahmed et al., 2025; Bhatt et al., 2025; Stroebl et al., 2024).

Our work belongs to a family of LLM safety methods that adopt a *proactive* approach: rather than waiting for a prompt X to elicit an unsafe response, we aim to predict the risk associated with the prompt in advance. Such safety methods estimate the probability of an unsafe event $\mathbb{P}(Y = 1 \mid X)$, where Y is a binary safety indicator (Perez et al., 2022; Su et al., 2024; Xu et al., 2025). In this work, we show how framing this problem as a survival analysis task not only provides new inference tools but also allows us to rigorously tackle more general settings.

To understand the challenges, one can view the unsafe probability estimation as a process that independently generates multiple LLM responses for a given X, audits their safety, and computes the average of the resulting unsafe indicators. This process highlights two key limitations. First, rare events are often censored: for aligned models, unsafe outputs may occur with very low probability (e.g., 10^{-6}) and thus remain unobserved under reasonable sampling budgets $B \in \mathbb{N}$ (e.g., 10 LLM generations per prompt). In such cases, naively fitting a classifier to estimate the unsafe probability can misleadingly ignore the under-

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lying risk. Second, LLM generations are not always independent. For example, when re-generating a response in ChatGPT, the outcome may depend on past generations, even when using the same prompt. Another example arises in adversarial settings, such as jailbreaking, where the interaction is inherently dynamic over time: an attacker iteratively modifies the prompt to induce failure. In these cases, the probability of an unsafe event is time-dependent; consequently, standard estimators become invalid (Rao et al., 2024; Dickson et al., 2024).

To handle these limitations in a unified and rigorous way, we propose a fundamental change in perspective: rather than asking what is the probability that the model will fail, we ask *when the first failure will occur*. We introduce **time-to-unsafe-sampling**, defined as the number of sampling attempts required to elicit an unsafe response for a given prompt X.

In the simplest setting where the prompts and responses are sampled i.i.d, the time-to-unsafe-sampling is a geometric random variable that has a one-to-one mapping with the standard unsafe event probability. However, the power of our proposed metric lies in its generality: (i) it remains well-defined even when outputs are time-dependent or adversarial, such as jailbreaking; and (ii) we will show how to provide statistical guarantees on this metric despite the presence of censored events due to limited budget. Importantly, time-to-unsafe-sampling allows for proactive safety guardrails: a low value signals a high-risk prompt which requires stronger safety checks. Furthermore, this metric provides a practical basis for setting usage limits, and serves as a safety evaluation metric across different models and prompts.

Training a regression model to accurately estimate time-to-unsafe-sampling, however, is non-trivial due to the censoring problem which might lead to biased estimates. Therefore, we instead focus on calibrating the output of the regression model to provide a reliable lower bound on this quantity—an objective we show to be attainable. Denote by $\hat{L}(X_{\text{test}})$ a lower predictive bound (LPB) on the unknown time-to-unsafe-sampling T_{test} of a new test prompt X_{test}. We define the coverage rate of $\hat{L}(X_{\text{test}})$ as the probability that the true (unknown) time-to-unsafe-sampling T_{test} exceeds the LPB, i.e., $\mathbb{P}(T_{\text{test}} \geq \hat{L}(X_{\text{test}}))$. Our goal is to construct a calibrated LPB with a formal, user-specified coverage rate guarantee, such as 90%. Informally, this means that, with high probability, one can expect to generate at least $\hat{L}(X_{\text{test}})$ safe responses to the test prompt X_{test} before encountering an unsafe one across future test prompts.

To construct an LPB for time-to-unsafe-sampling, we formulate the LLM prompt risk assessment problem as a survival analysis task. In survival analysis, the goal is to predict a time-to-event—e.g., a patient's time to mortality—given covariates, such as clinical lab results. Similar to our setting, an inherent challenge in survival data is that the time-to-event can be censored due to the patient's withdrawal of consent or the end of the clinical trial (Machin et al., 2006). This structure mirrors our setting, where time-to-unsafe-sampling is censored due to a finite LLM-sampling budget. Leveraging this analogy, we adopt recent conformalized survival analysis methods, which provide a principled way to construct LPBs with rigorous, finite-sample coverage guarantees under censoring (Gui et al., 2024a; Candès et al., 2023).

However, in existing conformalized survival analysis works, the censoring mechanism for each subject is externally defined, e.g., by study design, and treated as fixed. Put simply, these methods do not intervene in how the data is collected. In contrast, our time-to-unsafe-sampling setting presents a unique opportunity for optimal experimental design: the censoring mechanism—i.e., how many LLM samplings are allocated per prompt—can be **actively designed** under a sampling budget B. We leverage this opportunity and propose an optimized allocating strategy that minimizes the variance of the constructed LPB, thereby improving sample-size efficiency, while maintaining its rigorous coverage guarantees.

In sum, our main contributions are as follows:

(1) We propose time-to-unsafe-sampling: a new safety measure that quantifies the risk of temporal, probabilistic failures at a fine-grained, per-prompt level (Section 2).

(2) We refine the method of Gui et al. (2024a), simplifying it to construct LPBs with a formal coverage guarantee under a known censoring mechanism (Section 4). Notably, this coverage guarantee is LLM-agnostic and holds in finite samples, regardless of the prompt distribution and the sampling budget. Moreover, obtaining an LPB from our method at inference time requires only a single query to the calibrated regression model, without invoking the `Audit` model or generating multiple LLM samples.

(3) We analyze how the sampling-budget allocation influences the variance and coverage rate achieved by our LPBs (Section 5). Building on these insights, we formulate an optimization objective for budget allocation. Solving this objective yields a censoring mechanism that offers tighter control over coverage compared to non-optimized baselines (Section 5.2).

(4) We validate our theory on simulated and real-data experiments. Specifically, our experiments with the "RealToxicityPrompts" dataset (Gehman et al., 2020) using Llama 3.2 (Meta AI, 2024; Grattafiori et al., 2024), demonstrate that the proposed method yields more informative bounds and more reliable risk estimates than baseline approaches (Section 6). Code is provided for reproducibility at http://github.com/giladfrid009/llm-survival/.

2 Problem Setup

Let $\mathcal{D}=\{X_i\}_{i=1}^n$ be a dataset of user prompts, drawn i.i.d. from some distribution P_X. Let $\mathcal{G}(x)$ denote an LLM. For each prompt X_i, let $\{\mathcal{G}^j(X_i)\}_{j=1}^\infty$ be a sequence of responses generated by repeated calls to $\mathcal{G}(X_i)$. We define the binary indicator $Y_i^j = \texttt{Audit}\big(X_i, \mathcal{G}^j(X_i)\big)$, where $Y_i^j = 1$ if and only if the j-th response is labeled unsafe by the `Audit` function. The uncensored *time-to-unsafe-sampling* for prompt X_i is then

$$T_i = \min\big\{\, j \geq 1 : Y_i^j = 1 \big\}.$$

In the simplest case, conditional on X_i, the sequence $\{Y_i^j\}$ is i.i.d. Bernoulli with unsafe sampling probability $p(X_i) := \mathbb{P}(Y_i^j = 1 \mid X_i)$, implying $T_i \mid X_i \sim \text{Geom}\big(p(X_i)\big)$. In practice, however, successive generations may be dependent; for instance, through iterative self-reflective prompting (Madaan et al., 2023; Shinn et al., 2023), jailbreak attacks (Chao et al., 2025), or LLM tool use where tool availability or behavior changes over time (Wang et al., 2023). Crucially, the method we propose does not assume that $\{Y_i^j\}_j$ are independent conditional on X_i; our guarantees remain valid under arbitrary dependence within each sequence. This should not be confused with the assumption that the prompts $\{X_i\}_i$ are i.i.d., which is required for our theoretical analysis.

Given a user-specified coverage level $1-\alpha \in (0,1)$, and a tolerance level $\delta \in (0,1)$, our goal is to construct an LPB $\hat{L}$ that satisfies

$$\mathbb{P}\big(T_{\text{test}} \geq \hat{L}(X_{\text{test}}) \mid \mathcal{D}\big) \geq 1-\alpha, \tag{1}$$

with probability at least $1-\delta$ over the randomness of the training data $\mathcal{D}$. The probability in (1) is taken over the distribution of a test point $(X_{\text{test}}, T_{\text{test}}) \sim P_{X,T}$. An LPB $\hat{L}(X_{\text{test}})$ satisfying this requirement is called a Probably Approximately Correct (PAC) LPB at level α with tolerance δ.

We follow standard split-conformal methodology (Vovk et al., 2005), and partition the index set $\{1,\ldots,n\}$ of the training prompts $\mathcal{D}$ into a *proper training set* $\mathcal{I}_1$ and a holdout *calibration set* $\mathcal{I}_2$. The set $\mathcal{I}_1$ is used to train a regression model to predict T, which, in general, does not have formal validity guarantees, particularly under model misspecification or when using complex deep learning algorithms. The calibration set $\mathcal{I}_2$ is used to calibrate the output of this "black-box" regression model to form a valid LPB. Our focus in this work is on the calibration stage.

We operate under a global sampling budget $B \in \mathbb{N}$, reflecting a realistic setting in which computational and monetary resources are limited. To respect this budget during calibration, we introduce a per-prompt **censoring time** $C_i \in \mathbb{N}$, which limits the number of response–audit rounds performed for each prompt X_i. Formally, we require that

$$\mathbb{E}\Big[\sum_{i\in\mathcal{I}_2} C_i \;\Big|\; \{X_i\}_{i\in\mathcal{I}_2}\Big] \leq B. \tag{2}$$

In words, the above ensures that the expected number of response–audit rounds does not exceed the overall budget B.[1]

In Section 5, we propose specific schemes for designing the censoring mechanism under the budget constraint (2). This results in a calibration dataset of the form $\{X_i, C_i, \tilde{T}_i\}_{i\in\mathcal{I}_2}$, where $\tilde{T}_i = \min\big(T_i, C_i\big)$ denotes the censored time-to-unsafe-sampling. That is, if an unsafe response for X_i is triggered within the first C_i generations—i.e., among $Y_i^1, Y_i^2, \ldots, Y_i^{C_i}$—then $\tilde{T}_i = T_i$; otherwise, the time-to-unsafe-sampling is censored at C_i, and we have $\tilde{T}_i = C_i$.

In addition to satisfying the global budget constraint (2), we design the censoring mechanism to satisfy a widely adopted assumption in the survival literature that the censoring time C is conditionally independent of the survival time T given the covariates X, which provides a theoretical guarantee on the LPB we construct:

Assumption 2.1 (Conditionally Independent Censoring (Kalbfleisch and Prentice, 2011))**.** $C \perp\!\!\!\perp T \mid X$.

This assumption means we cannot dynamically adjust the sampling budget based on the generated LLM responses, as doing so would induce dependencies between T_i and C_i, invalidating the statistical calibration. However, it allows us to optimize the allocation C_i based on the prompt X_i before sampling begins–a key degree of freedom we exploit in Section 5.2.

3 Related Work

In this section, we present the conformalized survival analysis algorithm of Gui et al. (2024a), called

[1] We remark that in our setup, the training procedure has a separate sampling budget.

`Adaptive-T`, which constructs a valid LPB in the sense of (1) under Assumption 2.1. Additional related work is discussed in Appendix A.

Given calibration data $\{X_i, C_i, \tilde{T}_i\}_{i\in\mathcal{I}_2}$, `Adaptive-T` constructs a PAC-type LPB $\hat{L}$ by calibrating a pre-trained quantile regression model. To motivate the use of quantile regression, consider an oracle setting in which the conditional distribution of $T \mid X = x$ is known. Let $q_\tau(x)$ denote the true τ-th conditional quantile of $T \mid X = x$. In this case, the oracle LPB at coverage level $1-\alpha$ for a new test prompt X_{test} is simply $\hat{L}(X_{\text{test}}) = q_\alpha(X_{\text{test}})$, which achieves the desired coverage by definition.

Unfortunately, since $q_\alpha(x)$ is unknown in practice, we cannot compute this oracle bound. Instead, we can estimate the conditional quantile function, denoted by $\hat{q}_\tau(x)$, and then obtain a naive plug-in LPB: $\hat{L}(X_{\text{test}}) = \hat{q}_\alpha(X_{\text{test}})$. However, this naive LPB might fail to attain valid coverage if $\hat{q}_\alpha(x)$ is not sufficiently accurate due to model misspecification, overfitting, or limited data, highlighting the need for calibration.

The `Adaptive-T` method addresses this limitation by finding a calibrated quantile level $\hat{\tau}^{\text{adapt}}$ such that $\hat{L}(X_{\text{test}}) = \hat{q}_{\hat{\tau}^{\text{adapt}}}(X_{\text{test}})$ approximately satisfies (1) in a PAC sense. At a high level, this calibration is done by estimating the miscoverage probability $\mathbb{P}\big(T < \hat{q}_\tau(X)\big)$ for each candidate τ using the held-out calibration set $\mathcal{I}_2$. Then, this method proceeds by choosing the largest τ for which all $\tau' < \tau$ have estimated miscoverage less than or equal to α.

In detail, for each level τ, the miscoverage estimator $\hat{\alpha}^{\text{adapt}}(\tau)$ proposed by Gui et al. (2024a) is given by

$$\hat{\alpha}^{\text{adapt}}(\tau) = \frac{\sum_{i\in\mathcal{I}_2} \hat{w}_\tau(X_i)\,\mathbb{I}\{\tilde{T}_i < \hat{q}_\tau(X_i) \le C_i\}}{\sum_{i\in\mathcal{I}_2} \hat{w}_\tau(X_i)\,\mathbb{I}\{\hat{q}_\tau(X_i) \le C_i\}}. \quad (3)$$

We pause to clarify the above expression. First, observe that the indicator $\mathbb{I}\{\tilde{T}_i < \hat{q}_\tau(X_i) \le C_i\}$ is a function of observed quantities. Second, under Assumption 2.1, Gui et al. (2024a) show that considering only samples for which $\hat{q}_\tau(X_i) \le C_i$ introduces a covariate shift. To account for this shift, each selected example is re-weighted by an "inverse-censoring" weight $\hat{w}_\tau(X_i)$ that approximates $1/\mathbb{P}\big[\hat{q}_\tau(X_i) \le C_i \mid X_i\big]$. Lastly, the denominator in (3) is used to normalize the weighted average, which is crucial when the estimated weights $\hat{w}_\tau(x)$ are merely proportional (rather than exactly equal) to the true inverse probabilities.

Finally, `Adaptive-T` defines the calibrated quantile level as $\hat{L}(x) = \hat{q}_{\hat{\tau}^{\text{adapt}}}(x)$, where

$$\hat{\tau}^{\text{adapt}} = \sup\Big\{\tau \in \mathcal{T} : \sup_{\tau'<\tau} \hat{\alpha}^{\text{adapt}}(\tau') \le \alpha\Big\}, \quad (4)$$

where $\mathcal{T}$ is the search space for τ. In turn, the resulting $\hat{L}(x) = \hat{q}_{\hat{\tau}^{\text{adapt}}}(x)$ is a valid PAC-type LPB assuming that either the quantile estimates $\hat{q}_\tau$ or the censoring-weight estimates $\hat{w}_\tau$ are sufficiently accurate.

4 Calibration with a known censoring mechanism

Recall that in our LLM prompt risk assessment setting, we design the sampling-budget allocation scheme. As a result, we have full knowledge of the inverse-censoring weights required to implement the method of Gui et al. (2024a) presented in the previous section. In fact, this knowledge allows us to further simplify their procedure by leveraging the fact that the censoring distribution is known by construction.

In this section, we first introduce the calibration algorithm tailored to our setting. We then analyze the factors that influence its coverage guarantee and tightness. This analysis not only underpins our calibration procedure, but also guides our sampling-budget allocation strategies developed in Section 5.

Formally, the inverse-censoring weight corresponding to each $\hat{q}_\tau$ is defined as

$$w_\tau(X_i) = 1/\mathbb{P}\big[\hat{q}_\tau(X_i) \le C_i \mid X_i\big]. \quad (5)$$

Using these weights, we estimate the empirical miscoverage rate at quantile level τ as a weighted average over the calibration set:

$$\hat{\alpha}(\tau) \;=\; \frac{1}{|\mathcal{I}_2|}\sum_{i\in\mathcal{I}_2} w_\tau(X_i)\,\mathbb{I}\big\{\tilde{T}_i < \hat{q}_\tau(X_i) \le C_i\big\}. \quad (6)$$

Note that this estimator differs from the one used in (3): it does not require additional normalization since our proposed censoring mechanisms provide the true weights $w_\tau(X_i)$, as explained in Section 5.

Armed with $\hat{\alpha}(\tau)$, we obtain calibrated LPB by following (4) and selecting the largest τ such that $\hat{\alpha}(\tau) \le \alpha$:

$$\hat{L}(x) = \hat{q}_{\hat{\tau}}(x), \quad \hat{\tau} = \sup\big\{\tau \in \mathcal{T} : \sup_{\tau'<\tau} \hat{\alpha}(\tau') \le \alpha\big\}, \quad (7)$$

where $\mathcal{T} = \big\{\sup_{\tau\in[0,1]}\{\hat{q}_\tau(X_i) \le \tilde{T}_i\} : i \in \mathcal{I}_2\big\} \cup \big\{\sup_{\tau\in[0,1]}\{\hat{q}_\tau(X_i) \le C_i\} : i \in \mathcal{I}_2\big\} \cup \{0\}$.

This procedure is summarized in Algorithm 1. The following informal theorem shows that the LPB $\hat{L}$ formulated in (7) holds a PAC-type coverage guarantee. In Appendix D.6 we state the formal version of this theorem and prove it by building on the theoretical framework developed by Gui et al. (2024a).

Theorem 4.1 (General validity, informal)**.** *Fix a tolerance level $\delta \in (0,1)$ and a miscoverage level $\tau \in$*